# AN INDEPENDENT EXACT VERIFICATION OF THE VARIATIONAL STEP BEHIND THE CLAIMED 2026 BOUNDS $H_1 \le 240$ AND $H_1 \le 236$, WITH A REMARK ON THE CAPS OF A 212 DRAFT

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ABSTRACT. Several 2026 preprints claim improvements of the bound $H_1 \le 246$ of Polymath8b for $H_1 = \liminf(p_{n+1} - p_n)$. Each combines equidistribution estimates for primes in arithmetic progressions (new, or adapted from Stadlmann's) with a *variational step*: a test function $F$ on a capped region of $\mathbb{R}^k$ whose sieve ratio exceeds a threshold. We recompute the variational step of three of these claims in exact rational arithmetic, with independent programs, and do not address their analytic input. Ratios are given in the Polymath scaling, where the threshold is 4. (i) For the support and parameters of Stadlmann ($k = 49$, claimed $H_1 \le 240$) we certify the capped ratio $4.0120389\ldots > 4$ with a test function of our own, and $4.0124198\ldots$ after optimising it on the capped region; two independent programs agree exactly. (ii) For the support shared by Kintali and by Song and Yue ($k = 48$, claimed $H_1 \le 236$) we recompute the published witness of Song and Yue: its cap-free mass and its capped numerator equal the paper's reference integers (655 and 715 digits) exactly, so their criterion holds: in their normalisation (threshold 1) $48J_T/I_0 = 1.0006649215907\ldots > 2001/2000$. The authors list the regeneration of the capped numerator as pending. The same computation reproduces Kintali's printed capped and cap-free ratios to 12 significant digits, and a test function of our own gives $4.0060814\ldots > 4$ on the same support. The capped numerator of the witness and both functionals for our own test function are confirmed by a second, independent exact program. (iii) For the draft of Charton et al. ($k = 45$, claimed $H_1 \le 212$) the criterion holds for the caps of its Table 3 ($4.0027351\ldots > 4$, two programs), whereas for the different caps stated in its Lemma 9.1 no polynomial in an 846-dimensional space of the size the authors use reaches 3.99 (ball arithmetic). This locates an inconsistency in the draft; it is not a refutation. None of this verifies the new equidistribution estimates, on which every one of these bounds rests, and none of it bears on the twin prime conjecture, which remains open.

## 1. INTRODUCTION

**The reduction.** For an admissible $k$-tuple $\mathcal{H} = \{h_1 < \cdots < h_k\}$, let DHL$[k, 2]$ denote the statement that infinitely many translates $n + \mathcal{H}$ contain at least two primes. If DHL$[k, 2]$ holds and $\mathcal{H}$ is an admissible $k$-tuple of diameter $D$, then $H_1 \le D$. Maynard's method [4] proves DHL$[k, 2]$ from a level of distribution and a symmetric test function $F$ whose ratio $kJ(F)/I(F)$ exceeds a threshold; Polymath8b [7] enlarged the region on which $F$ may live and obtained $H_1 \le 246$ with $k = 50$. The 2026 preprints listed below enlarge the region further, to *capped* supports, and pay for this with equidistribution estimates for the moduli that arise: Stadlmann [9] proves new ones, Kintali [3] uses Stadlmann's [9, Prop. 3] with one branch made empty, Charton et al. [1] use Stadlmann's Type I, II and III estimates for their own family of moduli, and Song and Yue [8] prove their own.

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Prepared with the assistance of Claude (Anthropic). All programs, certificates and Lean sources are in the repository https://github.com/jerryckim16/twin-primes; a living companion paper with full details is at https://jerryckim16.github.io/twin-primes/.

**The claims examined.**

| claim | $k$ | source | what we check |
|---|---|---|---|
| $H_1 \le 240$ | 49 | Stadlmann [9] | variational criterion |
| $H_1 \le 236$ | 48 | Kintali [3]; Song–Yue [8] | variational criterion; published witness |
| $H_1 \le 212$ | 45 | Charton et al. [1] | variational criterion; caps |

Each claim has two parts: an *analytic* part (an equidistribution theorem for the relevant family of moduli) and a *variational* part (a function $F$ on the capped support with ratio above the threshold). We check only the variational part, and the combinatorial fact that an admissible tuple of the stated diameter exists. Throughout we label statements COMPUTED (exact certificate and a script that reproduces it), PROVED HERE, LEAN-CHECKED, LITERATURE (read in the source), or CLAIMED (asserted in a preprint and not verified by us).

**Results in brief.** Ratios in the Polymath scaling, where the threshold is 4:

| claim | variational ratio | status here |
|---|---|---|
| 240 | 4.0120389; optimised on $T$: 4.0124198 | COMPUTED, two programs |
| 236, witness of [8] | $48J_T/I_T = 4.0030988$; $48J_T/I_0 = 4.0026597$ $(= 4 \times 1.00066492$; SY criterion $> 2001/2000)$ | COMPUTED, two programs ($J_T$; $I_T$ one) |
| 236, our own $F$ | 4.0060814 | COMPUTED, two programs |
| 212, Table 3 caps | 4.0027351 | COMPUTED, two programs |
| 212, Lemma 9.1 caps | $< 3.99$ for all $F$ in an 846-dim. space | COMPUTED, ball arithm. |

**What this note does not do.** It does not verify the equidistribution estimates of [9], [3], [8] or [1]; all three claimed bounds remain conditional on them, and the established bound is still 246. Song and Yue [8, Remark 4.23] point to a missing hypothesis in one step of the proof of [9, Lemma 5]; we read that step and comment on it, and on which of the claims may depend on it, in Remark 10, without resolving it. Nothing here concerns the parity barrier or the twin prime conjecture.

## 2. The variational problem on a capped support

**Normalisations.** The sources work in coordinates $t$ in which the criterion reads $kJ/I > 1$ (Song and Yue use the slightly stronger $48J/I_0 > 2001/2000$, see below). We use the Polymath scaling $u = 4t$, in which the criterion reads $kJ/I > 4$: every length is multiplied by 4 (so $g = 4\delta$ and $C_m = 4B_m$ for the threshold $\delta$ and caps $B_m$ of [9], and the radius $U_{\mathrm{SY}} = 521/2000$ of [8] becomes $L = 1.042$), and every ratio $kJ/I$ is multiplied by 4. Values quoted from the sources have been converted, except where we say otherwise.

**Supports.** Let $\mathcal{R}_k = \{u \in [0,1]^k : \ u_1 + \cdots + u_k \le 1\}$ and write $L\mathcal{R}_k$ for its dilate. The one-band supports of [9] (LITERATURE: Definitions 1 and 5 and Proposition 1 of [9]) are described by a length $L$, a marginal radius $\tau$, a threshold $g$ and caps $C_1 \le C_2 \le \dots$, with $C_0 = 0$; with $\mathcal{L}(u) = \{i : u_i > g\}$ (the *large set*), $m(u) = |\mathcal{L}(u)|$ and $s(u) = \sum_{i \in \mathcal{L}(u)} u_i$,

$$T = \{u \in L\mathcal{R}_k : \ s(u) \le C_{m(u)}\}.$$

For a symmetric $F$ supported on $T$,

$$I_T(F) = \int_T F^2, \qquad J_T(F) = \int_{\tau\mathcal{R}_{k-1}} \left( \int F(u', u_k)\, du_k \right)^2 du',$$

and the variational criterion is $kJ_T(F)/I_T(F) > 4$. (This one-band form of $J_T$, with marginals over $\tau\mathcal{R}_{k-1}$, is our transcription of the criterion of [9, Prop. 1]; see [2, §5].) We write $I_0, J_0$

for the same functionals without caps, that is with $T$ replaced by $L\mathcal{R}_k$. In this scaling the parameters are as follows (LITERATURE, converted).

| | $k$ | $(L,\tau)$ | $g$ | caps $C_1, C_2, \dots$ |
|---|---|---|---|---|
| [9] | 49 | (1.042, 0.982) | 0.112 | 0.6, 0.6, 0.68, 0.68, … |
| [3], [8] | 48 | (1.042, 0.982) | 91/1250 | 1899/2500 for all $m$ |
| [1], Tab. 3 | 45 | (53/50, 249/250) | 41/625 | $4B_{1,m}$ (see below) |
| [1], L. 9.1 | 45 | (53/50, 249/250) | 41/625 | 0.62, 0.62, 0.68, 0.68, … |

For [1], Table 3 lists $B_{1,m} = \frac{1}{5000}(777, 794, 875, 917, 953, 983, 1016, 1042, 1063, 1081)$ for $m \le 10$ and $B_{1,m} = 1081/5000$ for $m \ge 11$, and $C_m = 4B_{1,m}$; Lemma 9.1 states $C_1 = C_2 = 31/50$ and $C_m = 17/25$ for $m \ge 3$. All four cap sets satisfy $g < C_m \le C_{m+1} \le C_m + g$ for $m \ge 1$ (checked by hand and in Python; the case of the Table 3 caps is also LEAN-CHECKED, `charton_caps_valid`). The inequality fails at $m = 0$ for every cap set, which is why Proposition 2 assumes it only for $m \ge 1$.

**Exact moments.** All integrals reduce to moments over simplices.

**Lemma 1** (PROVED HERE; elementary)**.** *Let $P_j = \sum_i u_i^j$, $X_1, \dots, X_k$ independent exponential variables of mean 1, and $E_k(b) = \mathbb{E}\prod_j P_j(X)^{b_j}$. Then, with $w(b) = \sum_j j b_j$,*

$$\int_{\mathcal{R}_k} (1-P_1)^A \prod_j P_j^{b_j}\, du = \frac{A!}{(k+A+w(b))!}\, E_k(b), \qquad \sum_b E_k(b)\frac{s^b}{b!} = \left(\sum_c w(c)!\,\frac{s^c}{c!}\right)^k.$$

*Proof.* The Dirichlet integral gives $\int_{\mathcal{R}_k}(1-P_1)^A u^\alpha\, du = A!\,\alpha!/(k+A+|\alpha|)!$ and $\alpha! = \mathbb{E}X^\alpha$; sum over the monomials of $\prod_j P_j^{b_j}$, all of degree $w(b)$. The generating function follows from independence. □

**The fibres of a capped support.** Let $R = L - |u'|$ on the fibre over $u' \in \mathbb{R}^{k-1}_{\ge 0}$, and call $u'$ *admissible* if $s(u') \le C_{m(u')}$.

**Proposition 2** (PROVED HERE; elementary)**.** *Let $C_0 = 0$ and suppose $C_{m+1} \le C_m + g$ for all $m \ge 1$. Then the fibre $\{t \ge 0 : (u', t) \in T\}$ is empty when $u'$ is not admissible and equals $[0, \theta(u')]$ otherwise, where*

$$\theta(u') = \min\big(R,\ \max(g,\ C_{m(u')+1} - s(u'))\big).$$

*Consequently, with $Q(u', x) = \int_0^x P(u', t)\, dt$ for a polynomial $P$,*

$$J_T(P\,1_T) = J_0(P) - \int_{\substack{|u'|\le\tau \\ u' \text{ not adm.}}} Q(u', R)^2 - \int_{\substack{|u'|\le\tau \\ u' \text{ adm., } \theta<R}} \big(Q(u', R)^2 - Q(u', \theta)^2\big),$$

*and $I_T(P\,1_T) = I_0(P) - \int_{L\mathcal{R}_k\setminus T} P^2$.*

*Proof.* If $t \le g$ the large set of $(u', t)$ is $\mathcal{L}(u')$, so $(u', t) \in T$ if and only if $u'$ is admissible and $t \le R$. If $t > g$ the large set gains the last coordinate and the condition is $s(u') + t \le C_{m(u')+1}$. If $m(u') = 0$ then $s(u') = 0 = C_0$, so $u'$ is admissible. If $m(u') \ge 1$, the hypothesis $C_{m+1} \le C_m + g$ with $t > g$ forces $s(u') < C_{m(u')}$. So in either case a non-admissible $u'$ has an empty fibre, and for admissible $u'$ the fibre is $[0, \min(R, g)] \cup (g, \min(R, C_{m+1} - s)] = [0, \theta]$. The marginal of $P\,1_T$ is $Q(u', \theta)$ on admissible $u'$ and 0 otherwise; subtract from the cap-free marginal $Q(u', R)$. □

By symmetry each correction is a finite signed sum, over the size $m$ of the large set and over $j \ge 0$ further coordinates forced above $g$ (inclusion and exclusion for "all other coordinates are at most $g$"), of integrals of polynomials over polygons in two radial variables; so

every term is an exact rational number. The ranges that occur are short. In Lean (LEAN-CHECKED, file `lean/TwinPrimes/Caps.lean`): at most 9 coordinates of a point of $L\mathcal{R}_{49}$ exceed $g$ for [9] (`stadlmann_large_count`), at most 10 large coordinates of a point of $T$ for [3] (`kintali_large_count`), and at most 16 coordinates of a point of $L\mathcal{R}_{45}$ exceed $g$ for [1] (`charton_large_count`); and the inclusion–exclusion over further coordinates stops at $j \le 3$, 3, 6 respectively (`stadlmann_extra_le_three`, `kintali_extra_le_three`, `charton_extra_le_six`). The correction pieces themselves include non-admissible bases, whose size $m$ is bounded only by the total length (up to about 14 for [3]). The reduction to two radial variables is written out in [2, §5].

## 3. Method

**Test spaces.** For $D \ge 0$ let $V_D$ be the space of symmetric polynomials spanned by $(L - P_1)^a \prod_{i \ge 1} P_{2i}^{b_i}$ with $a + \sum_i 2ib_i \le D$; we call $D$ the *degree*. Its dimension is 707, 846 and 1041 for $D = 20$, 21 and 22. A "cap-free optimiser of degree $D$" is the top generalised eigenvector of $(kJ_0, I_0)$ on $V_D$.

**Two independent exact evaluators.** Given a polynomial $P$ with rational coefficients in some $V_D$, both programs return $I_T(P\,1_T)$ and $J_T(P\,1_T)$ as exact rationals.

(1) `capped_exact.py` computes the cap-free values with Lemma 1 and subtracts the corrections of Proposition 2, piece by piece.
(2) `capped_check.py` was written separately from the definitions only, without access to the first program. It sums directly over the exact large set, with exponential-moment generating functions, and has no cap-free term (details in [2, §5]).

Both were tested independently: a Monte Carlo comparison at $k = 12$ with caps that remove almost half of $I$ and $J$; exact agreement of `capped_check.py` with brute-force polytope triangulation and symbolic integration for $k = 2, \dots, 5$; and exact agreement of the two programs on every production input reported as "two programs" below. A value counts as certified when both programs return the same two rationals.

**Optimising on the capped support, and ceilings.** Both functionals are quadratic forms, $I_T = c^\top G_I c$ and $J_T = c^\top G_J c$, in the coefficients $c$ of $P$. Running the chain of exact linear maps of `capped_check.py` backwards gives the Gram matrices at about the cost of one evaluation (`capped_gram.py`; PROVED HERE, elementary). We use them in two ways. First, to *find* a good vector (the top generalised eigenvector), which is then rounded and re-evaluated exactly by the evaluators; only that exact value is stated as a certificate. Second, to prove a *ceiling*: if $\sigma G_I - kG_J$ is positive definite, no $P$ in the space reaches the ratio $\sigma$. Positive definiteness is certified by recursive Schur complements in ball arithmetic, with the entrywise error of $G_J$ included in the balls. The weights behind $G_I$ and $G_J$ were checked against `capped_check.py` exactly at a uniformly random integer vector, which by the Schwartz–Zippel lemma rules out a discrepancy between the two quadratic forms except with probability below $2^{-31}$ for each.

## 4. Results

**The support of Stadlmann ($k = 49$, claimed $H_1 \le 240$).**

**Theorem 3** (COMPUTED; exact, two independent programs)**.** *For the parameters of* [9] *in the table above:*

(1) *for $P$ the cap-free optimiser in $V_{20}$ (707 coefficients),*

$$49 J_T(P1_T)/I_T(P1_T) = 4.0120389035\ldots;$$

(2) *for* $P^*$ *the top eigenvector of* $(49G_J, G_I)$ *in the same space, rounded to* 400 *significant bits per coefficient,*

$$49J_T(P^*1_T)/I_T(P^*1_T) = 4.0124198475\ldots.$$

In both cases the two evaluators agree on $I_T$ and $J_T$ as exact rationals. Without caps the same space gives 4.016176, so the caps cost 0.0041 for the truncated optimiser and 0.0038 for the optimised one: optimising on $T$ regains less than a tenth of the loss. Every point of $T$ has all coordinates at most 0.6 (LEAN-CHECKED, `stadlmann_coord_le`) and $T \subset L\mathcal{R}_{49}$; the dual-weight theorem of [2, §4], applied as in [2, §5] and depending only on these two facts, gives $\sup_F kJ_T/I_T \le 4.14$ over *all* $F$ (COMPUTED, interval arithmetic, `dual_eps_cert.certify_box`, 337 pieces). So the margin of the variational step of [9] lies in (0.0124, 0.14].

**Corollary 4** (conditional; the analytic input is CLAIMED)**.** *The criterion of* [9, Prop. 1] *holds for* $k = 49$ *with Stadlmann's support, caps and parameters. With the* LEAN-CHECKED *admissible* 49*-tuple of diameter* 240*,* $H_1 \le 240$ *follows from* [9, Prop. 1] *and his equidistribution theorem, which we have not verified.*

**The support of Kintali and of Song–Yue ($k = 48$, claimed $H_1 \le 236$).** Both preprints use the same support (the table above, called "D182" in [3]) and, by their descriptions, the same public witness of degree 25, originally constructed for $k = 50$ and evaluated at $k = 48$ without re-optimisation. Song and Yue print it as 1295 integer triples $(a, \alpha, c)$, with

$$P(z) = \sum c\,\left(\tfrac{26}{25} - \textstyle\sum_i z_i\right)^a m_\alpha(z), \qquad P_{\mathrm{SY}}(t) = P\Big(\tfrac{26}{25U_{\mathrm{SY}}}t\Big), \quad U_{\mathrm{SY}} = \tfrac{521}{2000},$$

$m_\alpha$ the monomial symmetric polynomials, and they print the reference integers $I_{\mathrm{ref}} = (98!)^2 I_0^*$ and $J_{\mathrm{ref}} = (99!)^2(26!)^2 J^*$. Here $I_0$ is the *cap-free* mass on $\{\sum_i t_i < U_{\mathrm{SY}}\}$, $J$ the capped numerator, both in their $t$-coordinates, and the stars denote integer coordinates $x = 10^4 t$. Their criterion is $48J/I_0 > 2001/2000$ (threshold 1 in their normalisation), which implies the capped criterion because $I_T \le I_0$ (LITERATURE, [8, §4.12, App. A, App. C]). They state that a fresh evaluation reproduces $I_{\mathrm{ref}}$, while fresh regeneration of $J_{\mathrm{ref}}$ from the table "remains incomplete".

We extracted the table from the PDF using font sizes to read the exponents in the partitions. All 1295 rows are present, every $\alpha$ is nondecreasing with even parts, and $\max(a + |\alpha|) = 25$, as stated. Since all parts are even, $P_{\mathrm{SY}}$, rewritten in $u = 4t$, is exactly a polynomial in $L - P_1$ and the even power sums $P_2, \ldots, P_{24}$ (1311 terms); the conversion was checked exactly at random rational points.

**Theorem 5** (COMPUTED, exact rational arithmetic; items (1) and (2) each by two independent programs)**.** *For the witness* $P_{\mathrm{SY}}$ *of* [8] *and the support above, with* $I_0$ *and* $J_T$ *computed in the* $t$*-coordinates of* [8]*:*

(1) $(98!)^2\,10^{192}\,5^{150}\,521^{50}\,I_0 = I_{\mathrm{ref}}$ *exactly (an integer with* 655 *digits);*
(2) $(99!)^2(26!)^2\,10^{196}\,5^{150}\,521^{50}\,J_T = J_{\mathrm{ref}}$ *exactly (an integer with* 715 *digits), where* $J_T$ *is the capped numerator;*
(3) *hence, in the normalisation of* [8] *(threshold* 1*),* $48J_T/I_0 = 1.0006649215907\ldots > 2001/2000$*. In the Polymath scaling (threshold* 4*) the same ratio is* $4.00265968636\ldots$*; with the capped denominator* $48J_T/I_T = 4.0030988047\ldots$*, and without caps* $48J_0/I_0 = 4.0036348167\ldots$*.*

The factors $10^{192}5^{150}521^{50}$ and $10^{196}5^{150}521^{50}$ come from the change to integer coordinates when the polynomial is scaled by $65125^{25}$ (our reading of [8, Eq. (71)]; $65125 = 25 \cdot 2605$), so items (1) and (2) are equalities of integers, not numerical agreement. Item (1) was computed by `songyue_I0.py`, written for this purpose, which works in the monomial basis of the table, integrating each product $m_\alpha m_\beta$ by a sum over partial matchings of parts, and does not

use the power-sum conversion; `songyue_J0.py` computes $J_0$ the same way. Item (2) was computed by `capped_check.py` on the converted polynomial (56 minutes on four cores). Kintali [3] prints, in the normalisation with threshold 1, $48J_T/I_T = 1.000774701187468\ldots$ and $48J_0/I_0 = 1.000908704175987\ldots$, and the ratios $I_T/I_0 = 0.99989030538\ldots$ and $J_T/J_0 = 0.99975643873\ldots$; ours agree with all four to 12 significant digits; for example

$$4 \times 1.000774701187468 = 4.003098804749\ldots,$$

and our capped ratio is $4.0030988047499\ldots$.

*Second program (version 0.3).* The second evaluator, `capped_exact.py` (cap-free value minus 126 inclusion–exclusion correction pieces, each an exact rational; about 32 core-hours), gives the same $J_T$ as `capped_check.py` as an exact rational, so item (2) holds by two independent programs; its cap-free values equal $I_0$ and $J_0$ of the monomial-basis programs `songyue_I0.py` and `songyue_J0.py` exactly, so item (1) does too (see `code/songyue_compare2.py` and `code/cap6/compare2_sy236.json`). The capped denominator $I_T$, which enters only the ratio $48J_T/I_T$ and not the criterion of [8], rests on `capped_check.py` alone. Proposition 6 below is also confirmed by both programs (version 0.4).

**Proposition 6** (Computed, exact rational arithmetic, two independent programs)**.** *For our own cap-free optimiser in $V_{22}$ for this $(L,\tau)$ at $k = 48$ (1041 coefficients, rounded to 400 significant bits), $48J_T(P1_T)/I_T(P1_T) = 4.0060814208\ldots > 4$.*

`capped_check.py` and `capped_exact.py` (176 correction pieces, 15.5 core-hours) give the same $I_T$ and $J_T$ as exact rationals (`code/cap6/compare_kin48_D22r.json`). Here the caps cost only 0.00033 (the cap-free value is 4.0064115). That our value exceeds the witness's 4.0031 is expected, since the witness was optimised for $k = 50$ and transported, while ours is optimised for $k = 48$.

**Corollary 7** (conditional; the analytic input is Claimed)**.** *For the published witness, the variational criteria of both [3] ($48J_T/I_T > 4$) and [8] ($48J_T/I_0 > 2001/2000$) hold at $k = 48$. For our independent test function only the capped criterion $48J_T/I_T > 4$ was computed, and it holds (two programs). With the* Lean-checked *admissible 48-tuple of diameter 236, $H_1 \le 236$ follows from their equidistribution input, which we have not verified.*

**The draft of Charton et al. ($k = 45$, claimed $H_1 \le 212$).**

**Proposition 8** (Computed; exact, two independent programs)**.** *With the caps of [1, Table 3] and $P$ our cap-free optimiser in $V_{21}$ (846 coefficients, the dimension of the space $\mathcal{B}_{21}$ of [1, Thm. 11.1]), $45J_T(P1_T)/I_T(P1_T) = 4.0027351128\ldots > 4$.*

The authors report 4.00438 with their own optimised function (Claimed); our value is lower, as expected for a function optimised without caps, and already above 4. The draft, however, also states caps in its Lemma 9.1, namely $C_1 = C_2 = 31/50$ and $C_m = 17/25$ for $m \ge 3$, which are smaller than those of Table 3 for every $m \ge 1$. With these the conclusion changes.

**Proposition 9** (Computed, ball arithmetic)**.** *Let $T'$ be the support with the caps of Lemma 9.1 of [1] and $V = V_{21}$. Then $3.99\,G_I - 45\,G_J$ is positive definite on $V$, so*

$$\frac{45J_{T'}(P1_{T'})}{I_{T'}(P1_{T'})} < 3.99 \qquad \text{for every nonzero } P \in V.$$

The weaker statement with 4 in place of 3.99 was certified twice, at 5000 bits (blocks halved) and at 6500 bits (blocks split $1:2$); the statement with 3.99 was certified once, at 6500 bits. Both runs use the same saved weights, which agree exactly with `capped_check.py` at a random integer point. The best $P$ in $V$ gives about 3.98555 (floating point, from the Gram matrices). So within $V$, a space of the size the authors use, the caps of Lemma 9.1 cannot give their

4.00438, which is consistent with the caps of Table 3. We report this as a located inconsistency for the authors to resolve, not as an error in their result. We have not checked that their space coincides with ours beyond its dimension, and `capped_exact.py` has not yet re-evaluated the random point used for the weights.

## 5. Remarks

*Remark* 10 (on [9, Lemma 5]; Literature reading, not a result). Song and Yue [8, Remark 4.23] say that one step in the proof of [9, Lemma 5] lacks a hypothesis. In arXiv v1, Lemma 5 (a Type I estimate) takes $f = \alpha \star \beta$ with $\alpha$ an arbitrary coefficient sequence and $\beta$ smooth. For $\gamma$ in a short range above $1/2$ the proof applies Lemma 3 (a Type II estimate) "with the roles of $\alpha$ and $\beta$ swapped"; Lemma 3 requires its shorter factor to satisfy a Siegel–Walfisz condition, and after the swap that factor is $\alpha$, for which neither Lemma 5 nor the case of [9, Definition 9] where it is used assumes this. So the step reads as [8] describe it. A natural repair is to check that the sequences $\alpha$ arising in the decomposition have the property; we have not done so, and we do not know whether the conclusion is affected. Theorem 3 concerns only the variational step and is unaffected; the *bound* 240 depends on this input.

The question may also bear on the 236 of [3]: Kintali proves no new estimate and takes [9, Prop. 3], with its Type I, IIa, IIb and III branches and the Type IIc range made empty, as his analytic input (Literature). Whether his argument is affected depends on whether those branches pass through the step in question, which we have not traced. Song and Yue prove their own estimate [8, Thm. 4.18], via a Heath–Brown decomposition and Type IIa, IIb and III estimates [8, §§4.2–4.7]; we have not checked whether it uses an argument of the same kind ourselves; an AI audit made inside our project of the proof of [8, Thm. 4.18] (one auditor per section, not expert review) reports that the swapped-factor step does not occur there, and finds one repairable mis-statement, a gcd factor in the $\mathrm{Kl}_3$ correlation bound (49) of [8], inherited from [6, Lemma 6.25], that changes no exponent ([2], Remark on the audit; files in `notes/audit-sy236-thm418/`). Charton et al. [1] list the same five Type I, II and III estimates as [9], Lemma 5 among them, so the same question applies to their draft. Theorem 5 and Proposition 8 are unaffected either way.

*Remark* 11 (one-sided certificates; Proved here, Bonferroni). Only a lower bound for $J_T$ and an upper bound for $I_T$ are needed. In the inclusion–exclusion over the coordinates forced above $g$, every piece of $\int_V P^2$, of $\mathrm{Corr}_1$, and of the two nonnegative parts $\int Q(u', R)^2$ and $\int Q(u', U)^2$ of $\mathrm{Corr}_2$ has a nonnegative integrand, so the Bonferroni inequalities allow truncation after level 2 (upper bounds) or level 1 (lower bounds), and no piece with three forced coordinates is needed ([2], Propositions on Bonferroni truncation). For the vector of Theorem 3(1) this gives $49 J_{\mathrm{low}}/I_{\mathrm{up}} = 4.012038868 > 4$, $3.5 \cdot 10^{-8}$ below the exact value, as exact rationals (`capped_exact.py --bonf`; `code/cap6/compare_capfree_D20_bonf.json`).

*Remark* 12 (the next claim, 186; Computed, two independent programs). The claimed $H_1 \le 186$ of OpenAI [5] uses a different criterion, on fragment configurations with Dickman-distributed coordinate totals, at $k = 40$. We recomputed the two main terms of its numerical companion from the transcribed formulas, in our own code: $I_H = 2.36853178533 \cdot 10^{-14}$ and $J_{\lambda,H} = 9.0248755655 \cdot 10^{-14}$, inside the published enclosure for $I_H$ and above the published lower bound for $J_{\lambda,H}$, so that $\rho^* J_{\lambda,H}/I_H = 1.000206087 > 500103/500000$ ([2], Proposition on the 186 main terms; `code/oai186_IH.py`, `code/oai186_J.py`; a second program in ball arithmetic, `code/oai186_check2.py`, agrees to $7 \cdot 10^{-17}$). The loss bound of the companion, which the criterion also needs, and its analytic inputs are not checked.

*Remark* 13 (what would change the picture). The variational steps examined here have margins of 0.003 to 0.012 above the threshold. They are therefore robust to rounding but not to a change of support: Proposition 9 shows that a modest change in the caps moves the best ratio in $V_{21}$ below 4. Any revision of the analytic input that shrinks the admissible caps should be re-checked against the variational step, which our programs can do in hours.

## 6. Reproducibility

All inputs, certificates and logs are in the repository, under `code/` and `code/cap6/`. The evaluators are `code/capped_exact.py` and `code/capped_check.py`; the Gram matrices and ceilings come from `code/capped_gram.py`; the Song–Yue witness and reference integers, as extracted, are `code/cap6/songyue236_witness.json` and `code/cap6/songyue236_appC.json`, with the conversion and comparison scripts `code/songyue_*.py`. Running times above are for four cores and 16 GB of memory. The admissible tuples of diameters 212, 236, 240 and 246 are checked by the Lean 4 kernel (core Lean, no Mathlib, no `sorry`; axioms `propext` and `Quot.sound` only), as are the counting lemmas used in Proposition 2. The companion paper [2] contains the full derivations and the history of each computation: the dual-weight ceilings in its §4, and the capped supports, the two evaluators and the Gram matrices in its §5 (section numbers of its version 0.9).

## Changes

**0.5:** (14 November 2026) Remark 11 (one-sided certificates without the three-coordinate pieces) and Remark 12 (the main terms of the 186 companion). Remark 10 now mentions the internal audit of [8, Thm. 4.18].

**0.4:** (14 November 2026) The second exact program confirms Proposition 6 (our own $k = 48$ test function): same $I_T$ and $J_T$ as exact rationals. The one-program qualifiers for it are removed from the abstract, the table, the proposition and the corollary.

**0.3:** (13 November 2026) The second exact program confirms the capped numerator of the witness of [8] (Theorem 5); the one-program qualifiers for it are removed from the abstract, the table, the theorem and the corollary.

**0.2:** (13 November 2026) Revised after an internal referee report (in the directory `referee/` of the repository). The normalisations are now stated (new paragraph in §2) and the 236 ratios in Theorem 5 and the results table are given in one scaling; the one-program status of the 236 capped numerators is marked in the table, the theorem and the corollary, and the corollary no longer claims the criterion of [8] for our own test function; the cap-free agreement of the second evaluator is attributed correctly; the 4.14 ceiling has its provenance; "certified twice" is corrected for the 3.99 ceiling, and the statements after Proposition 9 are restricted to $V_{21}$, with a conjecture about the drafting of [1] removed; Proposition 2 sets $C_0 = 0$ and assumes the cap condition for $m \geq 1$ only, with the case $m(u') = 0$ added to the proof; the Lean attributions name the theorems and what they bound; the test spaces $V_D$ are defined; overloaded symbols are renamed ($\mathcal{L}$, $\theta$, $U_{\mathrm{SY}}$, $I_0$, $J_0$); "$m \geq 4$" is corrected to "$m \geq 1$"; truncated digits are corrected; Remark 10 now discusses which 236 argument may depend on the step it describes.

**0.1:** (13 November 2026) First draft.

## References

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